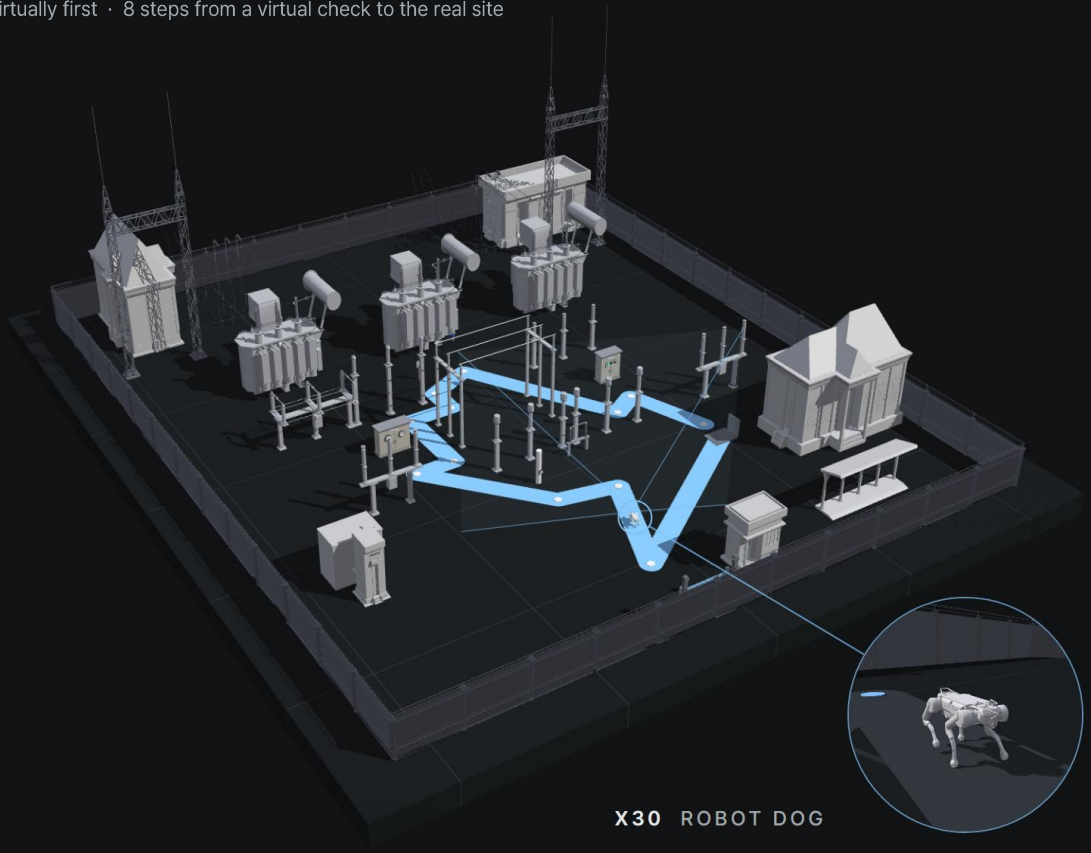
VALIDATE BEFORE INSTALLATION · SUBSTATION PATROL EDITION

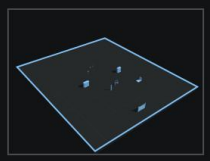
IntLearn Kinesis

Build a virtual substation, block by block

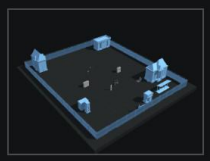
Site · buildings · equipment · robot dog · cameras · patrol route
Try the robot-dog patrol virtually first · 8 steps from a virtual check to the real site



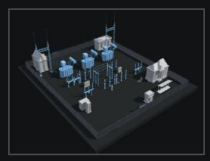
X30 ROBOT DOG



01 SITE



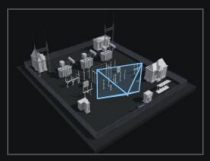
02 STRUCTURES



03 EQUIPMENT



04 ROBOT



05 CAMERAS



06 ROUTE

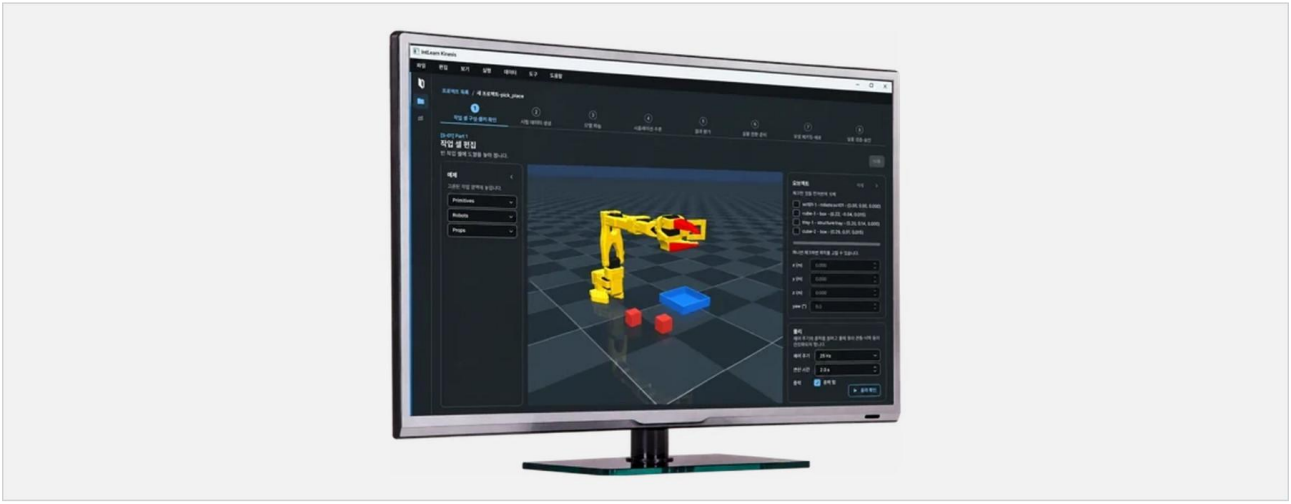
The IntLearn Kinesis example 'Outdoor substation patrol', built block by block into a virtual substation · X30 robot dog

02 — OVERVIEW

Stack a virtual substation, then run the robot dog's patrol first

Build robots, conveyors and cameras as a virtual cell, train on synthetic data (training images made virtually) instead of waiting for defects, and check performance before going on site. This brochure shows a robot dog patrolling a substation the same way. IntLearn Kinesis is new in 2026.

FIG. 01 SOFTWARE



IntLearn Kinesis software screen · example virtual cell (Korean UI)

8 steps	10	3	2026
From virtual to real	**Inspection targets**	**Robot dog models**	**New product**
Cell to real-world approval	Outdoor substation example	Spot · M20 · X30	Launched in 2026

Key functions From the product page

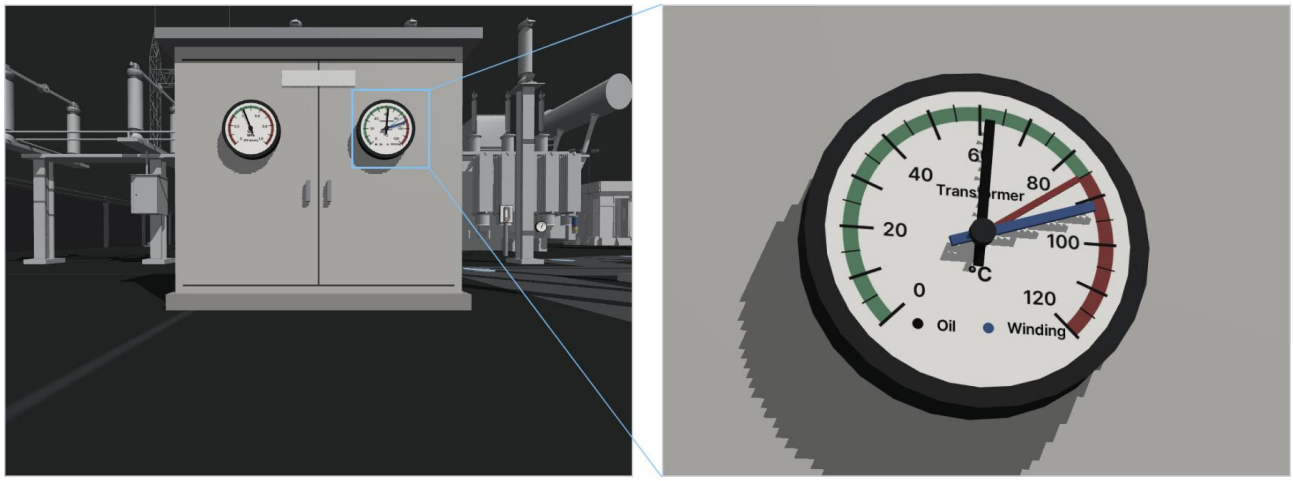
01	Virtual cell	Robot · conveyor · cameras, checked from four views
02	Synthetic data	Make defects instead of waiting for them
03	Training · simulated inference	Check performance before going on site
04	Real-world transfer · approval	Take what was validated virtually to the real line
05	Patent filed	Sim-to-Real: check it virtually, then apply it on the real line

03 — WHY VIRTUAL FIRST

Before a robot dog goes on site, what if you could run it first?

In a live substation it is hard to test in advance where the robot dog will walk and what its camera will see, and faults such as overheating or a needle out of range do not appear on demand. IntLearn Kinesis runs this test in a virtual substation first.

FIG. 02 ROBOT CAMERA · PTZ CLOSE-UP



Robot dog's front camera · stop at the gauge box

Same spot, zoomed in: the transformer thermometer

01

Walk the route first

Place the patrol stops (where the robot halts to inspect) and the route in the virtual substation; run it to the end.

02

See the camera view first

At each stop, the robot's camera and a zoomed view show whether every gauge can be read.

03

Don't wait for faults

Set needle positions and temperatures yourself and build a patrol with both normal and abnormal readings.

Before you install, check it in a virtual substation.

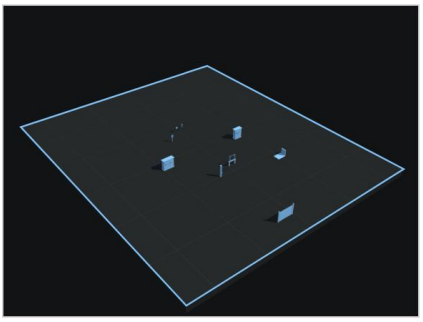
It starts by stacking the site, equipment, robot dog and cameras like blocks.

04 — BUILD LIKE BLOCKS

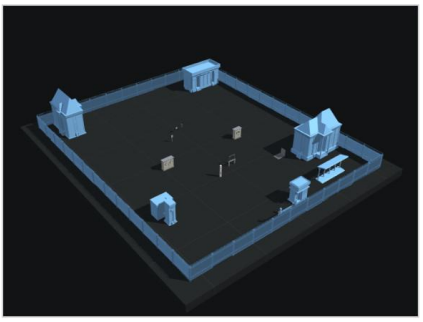
Build the virtual substation like stacking blocks

Add fences, buildings and equipment such as transformers to the site, place the robot dog and cameras, and link the patrol route. Pick from the parts list, drag, drop and check the layout from any side.

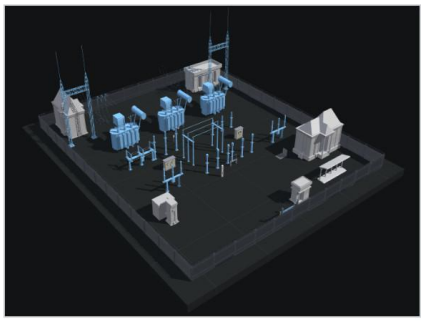
FIG. 03 ONE SCENE · SIX BLOCKS



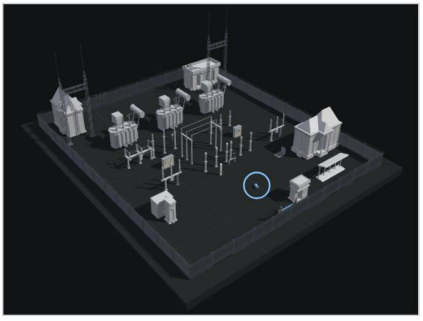
01 SITE Base
Ground · 10 targets · charging dock



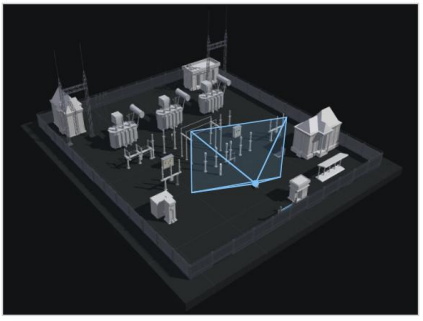
02 STRUCTURES Buildings
Fence · gate · guard booth · buildings



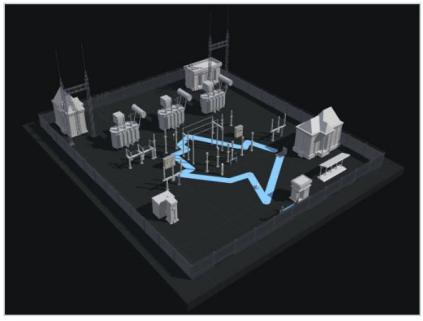
03 STRUCTURES Equipment
Transformer · switchgear · towers



04 ROBOTS Robot dog
One of X30 · M20 · Spot



05 CAMERAS Views
Robot front camera · overview



06 MISSION Route
Place stops; the route links them

The example 'Outdoor substation patrol' stacked one block at a time, same view · the newest block in blue

Parts list (asset palette)

Five groups to pick from in the work cell screen

PRIMITIVES	Shapes	Basic shapes: cubes, cylinders, cups, target markers
ROBOTS	Robots	Robot dogs Spot · M20 · X30, robot arms SO-101 · xArm7
PROPS	Props	Tabletop items: trays, conveyors, reject boxes
STRUCTURES	Structures	Fences, gates, main transformers, switchgear, towers
CAMERAS	Cameras	Fixed in one place, or mounted on a robot

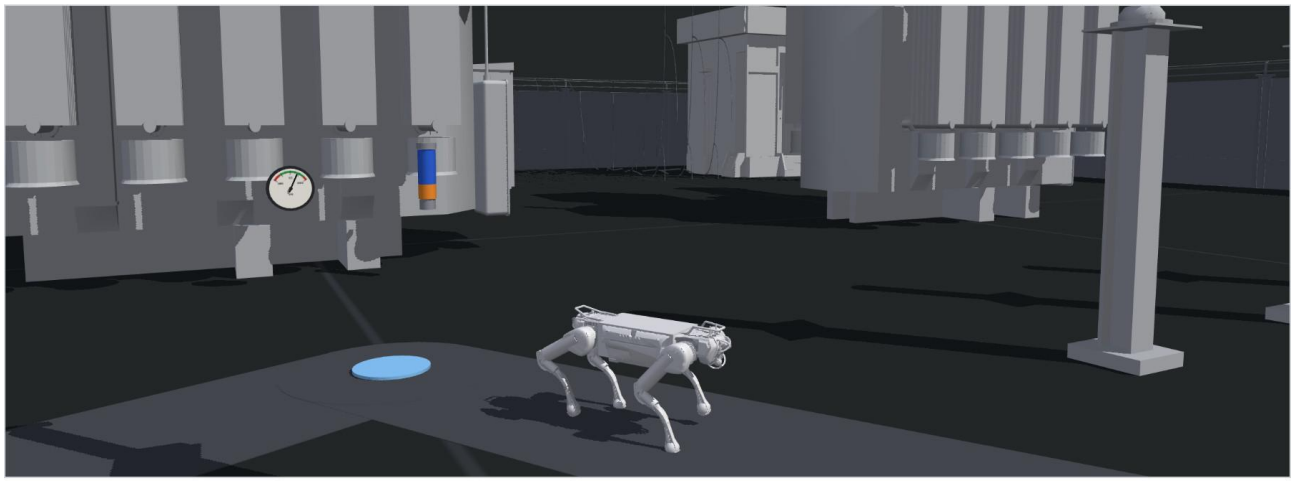
The example lays out the site (ground · targets · charging dock); the route follows the patrol stops and moves when you move them.

05 — QUADRUPED PATROL

A robot dog on virtual patrol

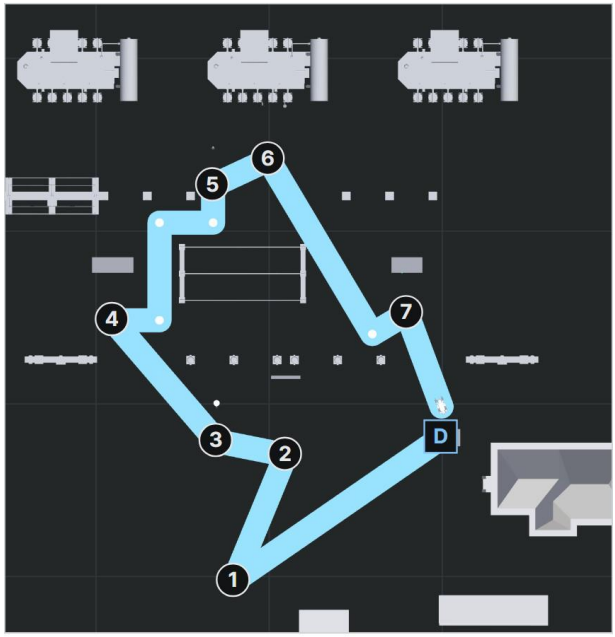
In the example 'Outdoor substation patrol', the robot dog leaves its charging dock, visits the door, disconnector, arrester, gauge box, oil level gauge, main transformer and indicator panel in turn, and returns to the dock. At each stop it halts and checks the target with its front camera.

FIG. 04 PATROL · X30



X30 heading to the next stop after the main transformer · behind it, the oil gauge and the breather

FIG. 05 ROUTE

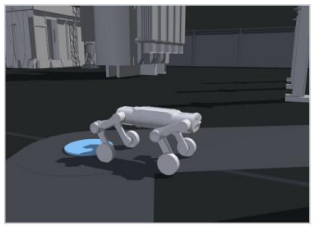


Patrol route · the virtual substation from above

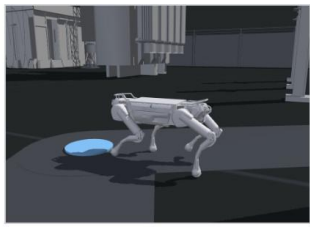
7 stops · 10 inspection targets

01	Door	Security check
02	Disconnector	Open or closed
03	Arrester	Operation count · leakage current
04	Gauge box	SF6 density · transformer temperature
05	Oil level gauge	Oil height (color band)
06	Main transformer	Hot spot · breather · oil gauge
07	Indicator panel	Lamp color
D	Charging dock	Start · return

FIG. 06 SWAP THE ROBOT



M20 wheeled quadruped



X30 walking quadruped

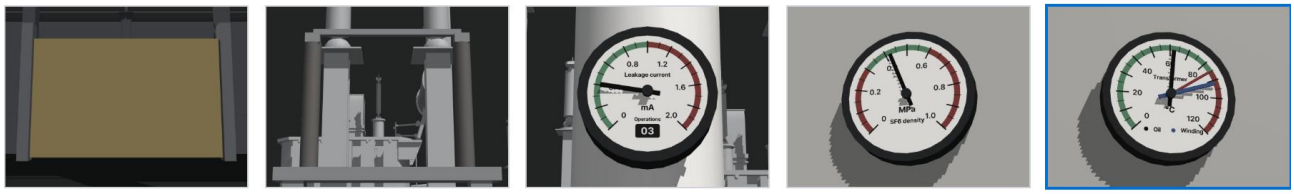
Same site, same patrol plan: only the robot block changes. X30 makes the same checks at the same stops as M20.

06 — INSPECTION · EVIDENCE

Don’t wait for faults: make them in a virtual substation

Set values on the virtual gauges and equipment, then run a patrol with normal and abnormal readings before anything goes wrong on site. Example alert: a 96 °C transformer hot spot (normal ≤85 °C).

FIG. 07 10 TARGETS · PTZ · THERMAL



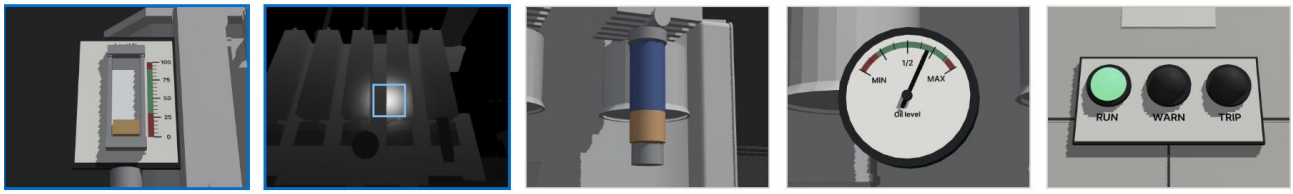
01 **Door**
Security check

02 **Disconnector**
Closed · normal

03 **Arrester counter**
0.4 mA (≤1.0)

04 **SF6 density**
0.42 MPa (0.3–0.6)

05 **Dual-needle gauge**
Winding 92 °C (≤85)



06 **Oil level gauge**
18 % (30–90)

07 **Hot spot**
96 °C (≤85)

08 **Breather**
Discolored 30 % (≤50)

09 **Oil gauge**
70 % (20–90)

10 **Indicators**
Run · normal

Zoomed views from the robot’s front camera at each stop · example set values (normal range) · blue = out of range

8 steps From virtual to real (Sim-to-Real)

From the cell to real-world approval

VIRTUAL Virtual substation · 01–05 **REAL** Real site · 06–08

01 Cell
Build the site, robot and cameras

02 Data
Make faults without waiting

03 Training
Train AI on virtual images

04 Inference
AI judges the virtual patrol

05 Evaluation
Check performance before going on site

06 Transfer prep
Get results ready for the real site

07 Deployment
Install on the real site

08 Real-world approval
Check and approve on site

IntLearn Kinesis: from virtual to real in 8 steps (Sim-to-Real)

Virtual first, the real site next.

The route and the AI tried in the virtual substation are checked and approved again on the real site.

07 — VIDEO · PLATFORM

Watch the robot dog’s patrol

All three Kinesis videos on the product page show a robot dog’s substation patrol. Point your phone camera at a QR code to watch.

Main video · 70 s · patrol simulation · English

What Does a Robot Dog Inspect at a Substation?

From a virtual temperature alert to doors, disconnectors, an arrester, a transformer and status indicators. Follow the route and see each target with its result card.

Series · 40 s · English

Build a Virtual Substation for Robot Dog Patrols

From the finished substation to equipment selection, drag-and-drop placement, position adjustments and camera views: the user builds the virtual environment.

Series · 35 s · model layout · English

Three Robot Dog Models, One Virtual Workspace

Add Spot, M20 and X30 to one virtual workspace, adjust their positions, then check the layout from above, beside and in front.

YouTube @intelliz

More on our YouTube channel

Works with these INTELLIZ products



IntCloud Optic

An AI optical module that captures defects so they can be seen



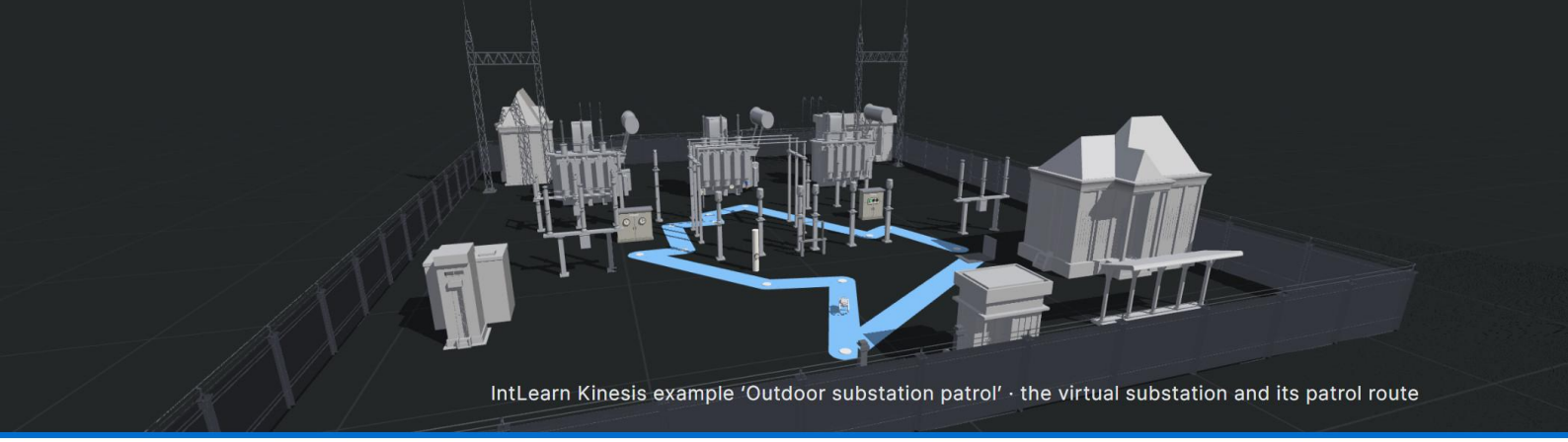
IntLearn Pro

Finds defects like an expert with 12 engines and modules (the most in Korea), directed in plain words



IntWise New in 2026

Say it, and the equipment is set; a veteran’s judgment is kept for the next operator



IntLearn Kinesis example ‘Outdoor substation patrol’ · the virtual substation and its patrol route

08 — CONTACT · NEXT STEPS

Considering a similar inspection?

Including the following in your inquiry helps us prepare our answer.

01	What you inspect: material, shape and size	02	The defects to find and your pass/fail criteria
03	Line speed and how products are conveyed	04	Space available for cameras and lighting
05	Where results need to go (PLC, reject unit, etc.)		

Next steps

Links at intelliz.net

Inspection guides	AI Visual Inspection vs Rule-Based Machine Vision	intelliz.net/en/guides/ai-visual-inspection-basics
Optics Lab	Can Line Scan or Area Scan Inspect It?	intelliz.net/en/solutions/optics-lab
Getting ready	AI Inspection Adoption Checklist	intelliz.net/en/tools/inspection-checklist
FAQ	Can performance be checked before installation?	intelliz.net/en/company/faq#before-install

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Product page

Try it

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Japan distributor BIT DEVICE Co., Ltd., 6-3-24-L430 Nishinakajima, Yodogawa-ku, Osaka, Japan

INTELLIZ

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Renderings: brochure styling made with the IntLearn Kinesis assembly engine; close-ups in reduced color.
Gauge labels are in English for this edition (the product example labels them in Korean).
Substation 3D models: kevinhalim9 · Gazebo Fuel (CC BY 4.0), meshes unchanged; placed, sized and colored here.
Robot models M20 · X30: DeepRoboticsLab (BSD-3-Clause)
Spot is a trademark of Boston Dynamics, M20 · X30 of DEEP Robotics; simulated scenes, not affiliated with either.